
BIM488 Introduction to Pattern Recognition

Speech Recognition

Outline

- Automatic Speech Recognition (ASR)
- Applications
- Human vs. Computer
- Issues in Speech Recognition
- ASR Approaches
- ASR Example

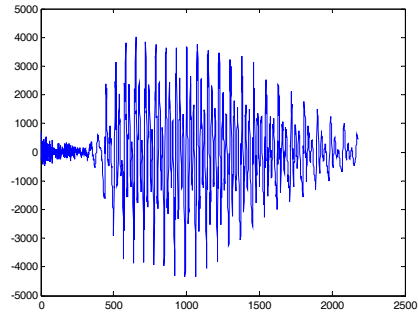
What is the task?

- Getting a computer to understand spoken language:
Automatic Speech Recognition
- By “understand” we might mean
 - React appropriately
 - Convert the input speech into another medium, e.g. text
 - etc.

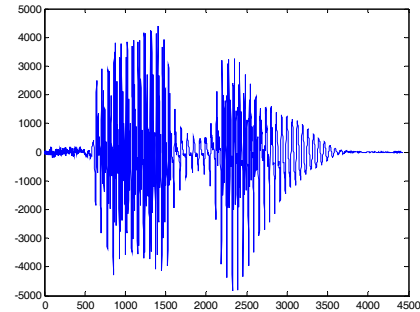
Applications

- Voice dialing
- Voice operated telephony systems
- Voice controlled devices
- Speech-to-Text converters
- Speaker recognition
- etc.

Samples of Speech Signal



'Two'



'Seven'

MATLAB: >> load('1007.mat')
 >> plot(wavedata)
 >> soundsc(wavedata)

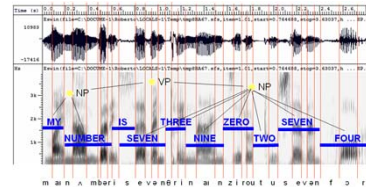
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How might computers do it?



- Digitization
- Acoustic analysis of the speech signal
- Linguistic interpretation



Speech recognition

Issues in Speech Recognition

- Digitization
 - Converting analogue signal into digital representation
- Signal processing
 - Separating speech from background noise
- Phonetics
 - Variability in human speech
- Phonology
 - Recognizing individual sound distinctions (similar phonemes)

Digitization

- Analogue to digital conversion
- Sampling and quantizing
- Use filters to measure energy levels for various points on the frequency spectrum
- Knowing the relative importance of different frequency bands (for speech) makes this process more efficient
- e.g. high frequency sounds are less informative, so can be sampled using a broader bandwidth (log scale)

Separating speech from background noise

- Noise cancelling microphones
 - Two mics, one facing speaker, the other facing away
 - Ambient noise is roughly same for both mics
- Knowing which bits of the signal relate to speech
 - Spectrograph analysis

Variability in individuals' speech

- Variation among speakers due to
 - Vocal range (f0, and pitch range – see later)
 - Voice quality (growl, whisper, physiological elements such as nasality, adenoidality, etc)
 - ACCENT !!! (especially vowel systems, but also consonants, allophones, etc.)
- Variation within speakers due to
 - Health, gender, emotional state
 - Ambient conditions
- Speech style: formal read vs spontaneous

Speaker-(in)dependent systems

- Speaker-dependent systems
 - Require “training” to “teach” the system your individual idiosyncracies
 - The more the merrier, but typically nowadays 5 or 10 minutes is enough
 - User asked to pronounce some key words which allow computer to infer details of the user’s accent and voice
 - Fortunately, languages are generally systematic
 - More robust
 - But less convenient
 - And obviously less portable
- Speaker-independent systems
 - Language coverage is reduced to compensate need to be flexible in phoneme identification
 - Clever compromise is to learn on the fly

(Dis)continuous speech

- Discontinuous speech much easier to recognize
 - Single words tend to be pronounced more clearly
- Continuous speech involves contextual coarticulation effects
 - Weak forms
 - Assimilation
 - Contractions

Approaches to ASR

- Template matching
- Knowledge-based (or rule-based) approach
- Statistical approach (machine learning)

Template-based approach

- Store examples of units (words, phonemes), then find the example that most closely fits the input
- Extract features from speech signal, then it's "just" a complex similarity matching problem, using solutions developed for all sorts of applications
- OK for discrete utterances, and a single user

Template-based approach

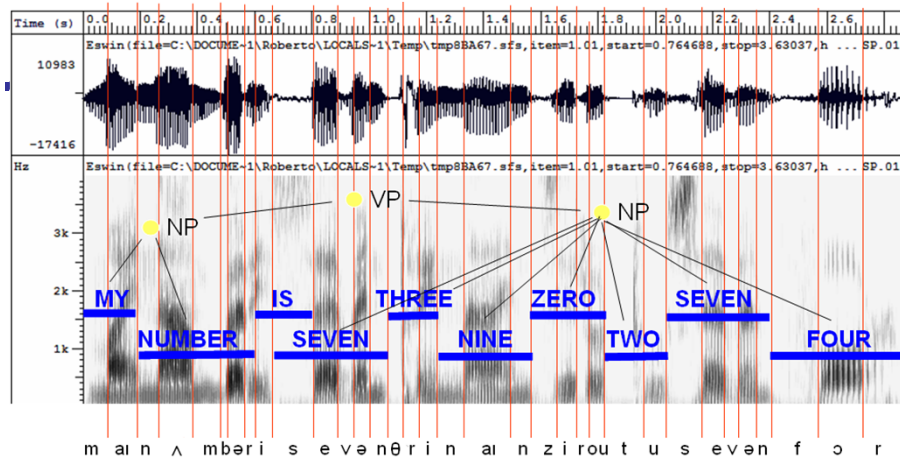
- Hard to distinguish very similar templates
- And quickly degrades when input differs from templates
- Therefore needs techniques to mitigate this degradation:
 - More subtle matching techniques
 - Multiple templates which are aggregated
- Taken together, these suggested ...

Rule-based approach

- Use knowledge of phonetics and linguistics to guide search process
- Templates are replaced by rules expressing everything (anything) that might help to decode:
 - Phonetics, phonology, phonotactics
 - Syntax
 - Pragmatics

Rule-based approach

- Typical approach is based on “blackboard” architecture:
 - At each decision point, lay out the possibilities
 - Apply rules to determine which sequences are permitted
- Poor performance due to
 - Difficulty to express rules
 - Difficulty to make rules interact
 - Difficulty to know how to improve the system



- Identify individual phonemes
- Identify words
- Identify sentence structure and/or meaning
- Interpret prosodic features (pitch, loudness, length)

Statistics-based approach

- Can be seen as extension of template-based approach, using more powerful mathematical and statistical tools
- Sometimes seen as “anti-linguistic” approach
 - Fred Jelinek (IBM, 1988): “Every time I fire a linguist my system improves”

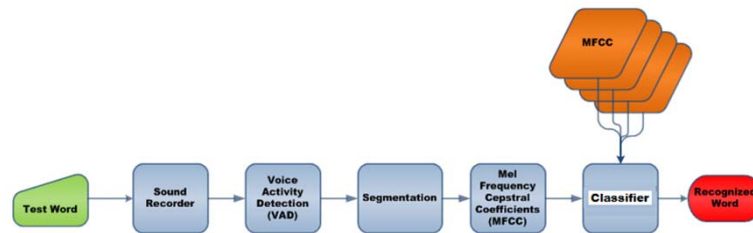
Statistics-based approach

- Collect a large corpus of transcribed speech recordings
- Train the computer to learn the correspondences (“machine learning”)
- At run time, apply statistical processes to search through the space of all possible solutions, and pick the statistically most likely one

ASR Example: Isolated Word Recognition

- Here, as an example, we will focus on isolated word recognition problem.
- We will talk about each step of the recognition process in detail.

ASR Example



Sound Recorder

- Analog to Digital conversion is carried out.
- Speech signal is now digital.
- Digital speech signal can now be processed.

Voice Activity Detection

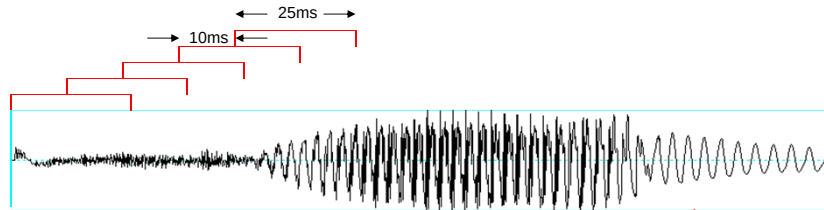
- Signal energy is compared with a pre-determined energy threshold.
- Thus, silence is discarded.

Segmentation (Windowing)

Words are parameterised on a frame-by-frame basis

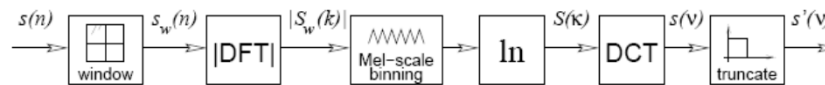
Choose frame length, over which speech remains reasonably stationary

Overlap frames e.g. 25ms frames, 10ms frame shift

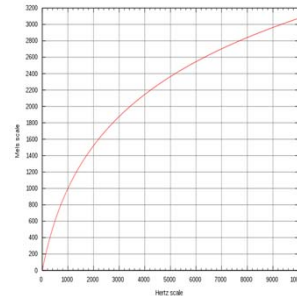


Computation of MFCC (Feature Extraction)

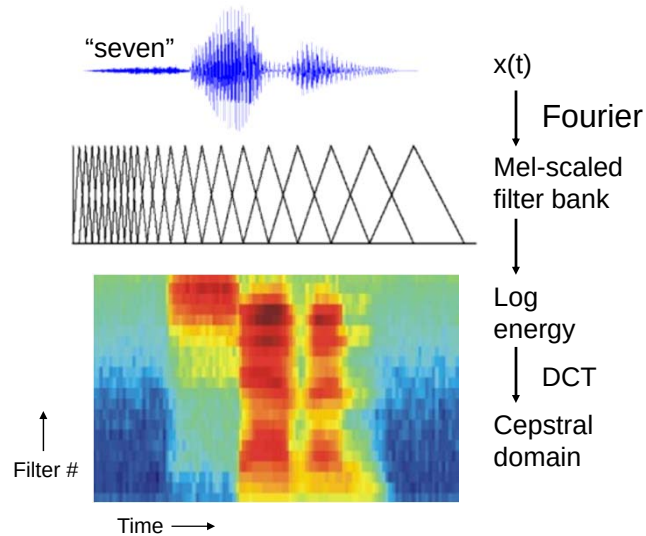
- Calculating Mel-frequency cepstral coefficients (MFCCs):



- MFCCs are coefficients of the short-term power spectrum of a sound, based on a linear cosine transform of a log power spectrum on a nonlinear *mel scale* of frequency.
- MFCCs are one of the most successful feature extraction approaches for speech data



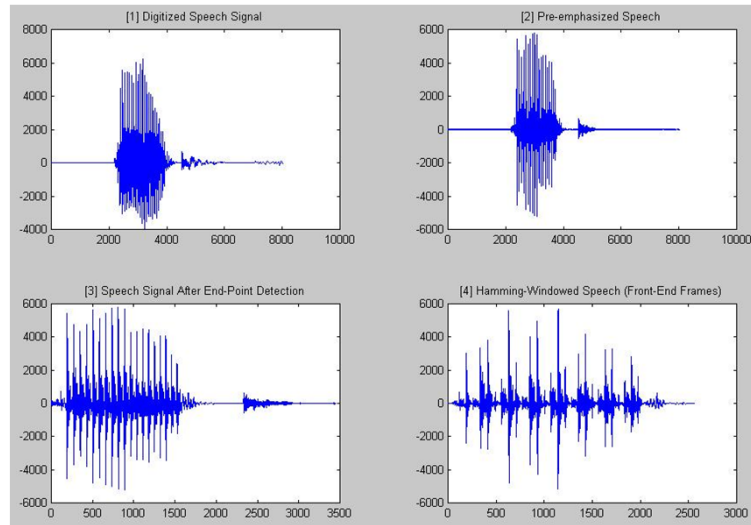
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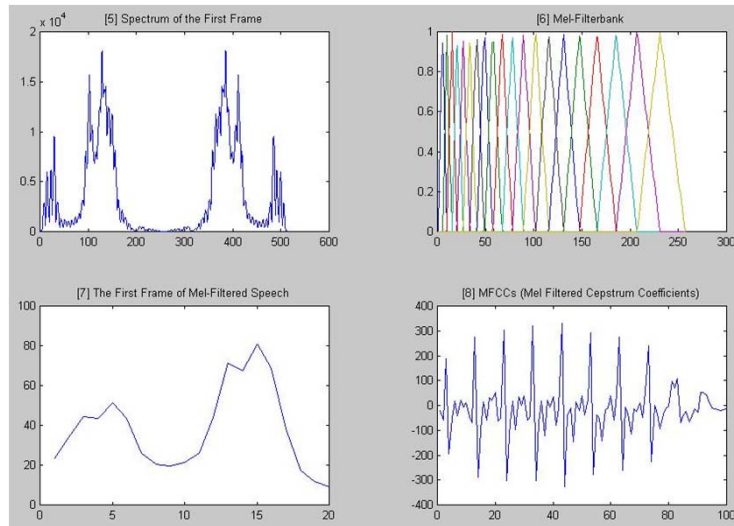
Classification

- Select a classification algorithm.
- Train your classifier using the training data.
- Then, start classifying unknown speech data.

Process Summary



Process Summary



Summary

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- Applications
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References

- S. Theodoridis and K. Koutroumbas, *Pattern Recognition* (4th Edition), Academic Press, 2009.
- Omid Talakoub, Astrid Yi, “Implementing a Speech Recognition System on a GPU using CUDA”.
- “Automatic Speech Recognition”, Informatics, The University of Manchester.

